45. (i) Half reaction are

$$\overset{+7}{M}nO_4^- \longrightarrow Mn^{2+} \quad \text{..... (1)}$$

and $\quad Sn^{2+} \longrightarrow Sn^{4+} \quad \text{..... (2)}$

(1) $\Rightarrow$ $\quad MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$

(2) $\Rightarrow$ $\quad Sn^{2+} \longrightarrow Sn^{4+} + 2e^-$

(1) $\times$ 2 $\Rightarrow$ $\quad 2MnO_4^- + 16H^+ + \cancel{10e^-} \longrightarrow 2Mn^{2+} + 8H_2O$

(2) $\times$ 5 $\Rightarrow$ $\quad 5\,Sn^{2+} \longrightarrow 5Sn^{4+} + \cancel{10e^-}$

$$\Rightarrow \quad 2MnO_4^- + 5Sn^{2+} + 16H^+ \longrightarrow 2Mn^{2+} + 5Sn^{4+} + 8H_2O$$

(ii)

$$\overset{+3}{C}_2O_4^{2-} \longrightarrow \overset{+4}{C}O_2 \quad \text{..... (1)}$$

$$\overset{+6}{Cr}_2O_7^{2-} \longrightarrow Cr^{3+} \quad \text{..... (2)}$$

(1) $\Rightarrow$ $\quad C_2O_4^{2-} \longrightarrow 2\,CO_2 + 2\,e^- \quad \text{..... (3)}$

$$Cr_2O_7^{2-} + 14\,H^+ \longrightarrow 2\,Cr^{3+} + 7\,H_2O \quad \text{..... (4)}$$

(3) $\times$ 3 $\Rightarrow$ $\quad 3\,C_2O_4^{2-} \longrightarrow 6\,CO_2 + \cancel{6\,e^-} \quad \text{..... (5)}$

$$Cr_2O_7^{2-} + 14\,H^+ + 6\,e^- \longrightarrow 2\,Cr^{3+} + 7\,H^2O \quad \text{..... (4)}$$

$$Cr_2O_7^{2-} + 3\,C_2O_4^{2-} + 14\,H^+ \longrightarrow 2\,Cr^{3+} + 6\,CO_2 + 7\,H_2O$$

(iii)

half reaction $\Rightarrow$ $\quad S_2O_3^{2-} \longrightarrow S_4O_6^{2-} \quad \text{..... (1)}$

$$I_2 \longrightarrow I^- \quad \text{..... (2)}$$

(1) $\Rightarrow$ $\quad 2S_2O_3^{2-} \longrightarrow S_4O_6^{2-} + \cancel{2e^-} \quad \text{..... (3)}$

(2) $\Rightarrow$ $\quad I_2 + \cancel{2e^-} \longrightarrow 2I^- \quad \text{..... (4)}$

(3) + (4) $\Rightarrow$ $\quad 2S_2O_3^{2-} + I_2 \longrightarrow S_4O_6^{2-} + 2I^-$

(iv) Half reactions are

$$\overset{0}{Z}n \longrightarrow Zn^{2+} \quad \text{..... (1)}$$

$$\overset{+5}{N}O_3^- \longrightarrow \overset{+2}{N}O \quad \text{..... (2)}$$

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